The Verification Deficit

Why string theory can't be verified — *yet* — measured, not just argued

The one honest sentence

People have said string theory is untestable for decades. We didn't discover *that*. We built the instrument that turns the criticism into a number.

The number

A claim like "this vacuum matches our universe" can only be *verified* with a certificate, and every certificate has a confidence radius:

$$\sqrt{\ln(1/\delta) / 2m}$$

where m is the number of observables that are simultaneously:

1. measured — we know the real answer,
2. computable — a solver can evaluate them, and
3. validated — the solver's error is calibrated.

For string theory today, m = 0. The three conditions never all hold at once. So the radius is:

$$\sqrt{\ln(1/\delta) / 0} = \infty$$

The certificate is vacuous — there is no error bound at all. That's not "hard." That's a measured, named, zero.

What we did *not* discover

The *why* was already known:

- the compactification → observables map is brutally hard to compute,
- the cosmological constant needs the unresolved uplift step,
- the measure (why this vacuum) is a mystery.

Smolin, Woit, and the whole field have known string theory is hard to test for thirty years. Nothing here is a new reason.

What we *did* build — the instrument

Three things the criticism never had:

1. The separation. Verification fails for one reason (m = 0); selection fails for a different reason (it's underdetermined). They were always conflated; now they're split, and each has its own name and its own fix.
2. The number. Not "it's really hard," but "the certificate is vacuous, and here is exactly what would close it."
3. The path. Each observable that gains a *validated* error model moves m from 0 toward ~10–19, shrinking the radius from ∞ toward ~31%. The deficit is a dial,

not a wall.

Why that matters for actually finding a solution

A criticism ends the conversation. An instrument tells you what to build next.

This one says, precisely: build the first validated oracle — a full spectrum computation calibrated against an exact case. That's a concrete, checkable, named task. Not a vibe. Not a debate. A job.

That's the real contribution, and it's a modest one: **we didn't find a new reason string theory is unverified — we quantified the known one, which is what makes it measurable, buildable, and improvable instead of just arguable.**

The name

The Verification Deficit.

The quantity: m (validated oracle outputs). The headline result: **the deficit is total — m = 0 today.**